

Supplemental Material for “Directional response jets of suppressed multiphoton events in a four-mode Fourier interferometer”

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I. PHASE HISTOGRAMS AND GENERATING COEFFICIENTS

Let $q^4 = 1$ and

$$L_j(q) = \sum_{k=0}^3 q^{jk} x_k. \quad (1)$$

For input $\mathbf{r}$ and output $\mathbf{s}$, define

$$\Phi_q(\mathbf{r}, \mathbf{s}) = \left(\prod_{k=0}^3 r_k! \right) [\mathbf{x}^{\mathbf{r}}] \prod_{j=0}^3 L_j(q)^{s_j}. \quad (2)$$

If $h_\ell(\mathbf{r}, \mathbf{s})$ counts labelled permanent paths with phase i^ℓ , then

$$\Phi_q(\mathbf{r}, \mathbf{s}) = \sum_{\ell=0}^3 h_\ell(\mathbf{r}, \mathbf{s}) q^\ell. \quad (3)$$

The four evaluations at $1, i, -1, -i$ determine the histogram by Fourier inversion.

II. FULL PROOF OF SECTORWISE RECIPROCITY

Fix nonnegative d, N , let $0 \leq a, s \leq d$, and let $0 \leq p, k \leq N$. We compare

$$\begin{aligned} \mathbf{r} &= (a, k, d-a, N-k), & \mathbf{s} &= (s, p, d-s, N-p), \\ \tilde{\mathbf{r}} &= (a, p, d-a, N-p), & \tilde{\mathbf{s}} &= (s, k, d-s, N-k). \end{aligned} \quad (4)$$

At $q = i$, set

$$E = x_0 + x_2, \quad T = x_0 - x_2, \quad S = x_1 + x_3, \quad D = x_1 - x_3. \quad (5)$$

The output generating polynomial factors as

$$\begin{aligned} A(E, S) &= (E + S)^s (E - S)^{d-s} = \sum_{m=0}^d \alpha_m E^{d-m} S^m, \\ B_p(T, D) &= (T + iD)^p (T - iD)^{N-p}. \end{aligned} \quad (6)$$

Use the unnormalized binary Krawtchouk coefficients

$$K_m(p; N) = \sum_t (-1)^t \binom{p}{t} \binom{N-p}{m-t}. \quad (7)$$

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Direct binomial expansion gives

$$[T^m]B_p = (-1)^{N-p-m} i^{N-m} K_m(p; N) D^{N-m}, \quad (8)$$

$$[x_0^a x_2^{d-a}] E^{d-m} T^m = (-1)^m K_a(m; d), \quad (9)$$

$$[x_1^k x_3^{N-k}] S^m D^{N-m} = (-1)^{N-m-k} K_k(m; N). \quad (10)$$

Let $C_{p,k}^{(a)}$ denote the coefficient before the input-factorial multiplier. Equations (8)–(10) yield

$$C_{p,k}^{(a)} = \sum_m \alpha_m (-1)^{N-p-m} i^{N-m} (-1)^m K_a(m; d) \times (-1)^{N-m-k} K_m(p; N) K_k(m; N). \quad (11)$$

All factors other than the final two Krawtchouk coefficients are unchanged under $p \leftrightarrow k$. The standard duality

$$\binom{N}{p} K_m(p; N) = \binom{N}{m} K_p(m; N) \quad (12)$$

therefore gives, term by term,

$$C_{p,k}^{(a)} = \frac{\binom{N}{k}}{\binom{N}{p}} C_{k,p}^{(a)}. \quad (13)$$

The input-factorial multipliers in Eq. (2) are

$$a!(d-a)!k!(N-k)! = \frac{a!(d-a)!N!}{\binom{N}{k}} \quad (14)$$

and the same expression with p replacing k . They cancel the ratio in Eq. (13).

At $q = -1$, the output polynomial is

$$(E + S)^d (E - S)^N, \quad (15)$$

independent of p . The requested monomials have total degree d in the even-indexed variables (x_0, x_2) and total degree N in the odd-indexed variables (x_1, x_3) , so only the $E^d S^N$ term contributes. Its two odd-mode coefficients have ratio $\binom{N}{k}/\binom{N}{p}$, canceled by the same factorials. At $q = 1$, both labelled values count all $n!$ permanent paths. The $q = -i$ equality is the complex conjugate of the $q = i$ equality. Fourier inversion proves equality of all four phase bins.

Ordinary input-output exchange, which holds because the canonical F_4 is symmetric, supplies the other two equalities in the fourfold statement. The product of input and output factorials is the same for the reciprocal pairs, so normalized Fock amplitudes are also equal.

A. Polynomial extension

For

$$H(z) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & z & -1 & -z \\ 1 & -1 & 1 & -1 \\ 1 & -z & -1 & z \end{pmatrix}, \quad (16)$$

the four generating forms are $E + S, T + zD, E - S, T - zD$. Replacing i by z in Eq. (8) contributes the common factor z^{N-m} and leaves Eq. (13) unchanged. Reciprocal permanents are thus identical polynomials in z . The passive-unitary interpretation is restricted to $|z| = 1$.

B. Boundary of the result

The histogram is defined in the canonical dephased gauge; arbitrary row and column phases need not preserve the four bins. Nor does parity alone imply an analogous theorem for F_8 . For example, the F_8 phase histograms for

$$\begin{aligned} (0, 0, 1, 0, 0, 1, 0, 0) &\longrightarrow (1, 0, 0, 1, 0, 0, 0, 0), \\ (0, 0, 1, 1, 0, 0, 0, 0) &\longrightarrow (1, 0, 0, 0, 0, 1, 0, 0) \end{aligned} \quad (17)$$

are unequal, although the odd-mode occupation subvectors are exchanged in the naive sense. The two binary sectors exposed by the order-four factorization, rather than parity alone, are essential.

III. DIRECTIONAL DERIVATIVES AND KRYLOV MOMENTS

For the unnormalized root matrix $2F_4$, write

$$Z_{\mathbf{r}, \mathbf{t}} = \text{per}(2F_4)[\mathbf{t}, \mathbf{r}], \quad D_{\mathbf{r}, \mathbf{s}} = 2^n \sqrt{\prod_j r_j! \prod_k s_k!}. \quad (18)$$

Second quantization maps X_{pq} and Y_{pq} to

$$\begin{aligned} \hat{X}_{pq} &= b_p^\dagger b_q + b_q^\dagger b_p, \\ \hat{Y}_{pq} &= -ib_p^\dagger b_q + ib_q^\dagger b_p. \end{aligned} \quad (19)$$

For a dark target, direct use of the ladder factors gives

$$\mathcal{A}'_X(0) = \frac{i}{D_{\mathbf{r}, \mathbf{s}}} (s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q} + s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q}), \quad (20)$$

$$\mathcal{A}'_Y(0) = \frac{1}{D_{\mathbf{r}, \mathbf{s}}} (s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q} - s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q}). \quad (21)$$

The square roots cancel against the normalization of the neighboring Fock state. At higher order, move one particle at a time in the output occupation graph. A move out of a mode with current occupation m contributes m ; for Y , it also contributes $+i$ or $-i$ according to the orientation. Summing the resulting Gaussian-integer weights times $Z_{\mathbf{r}, \mathbf{t}}$ gives the common-denominator numerator of $\langle \mathbf{s} | \hat{G}^j F_4 | \mathbf{r} \rangle$.

The two-mode sector reached from a target with $M = s_p + s_q$ has dimension $M + 1$. If the first $M + 1$ Krylov moments vanish, Cayley–Hamilton implies that the amplitude vanishes for the entire one-parameter rotation.

IV. COMPLETE EVENT-LEVEL RESPONSE TABLES

An entry $c\epsilon^{2j}$ means $P(\epsilon) = c\epsilon^{2j} + O(\epsilon^{2j+1})$.

A. Cyclic four-photon event

$$\mathbf{r} = (1, 1, 1, 1), \quad \mathbf{s} = (3, 1, 0, 0). \quad (22)$$

pair	X_{pq}	Y_{pq}
01	$3\epsilon^2/8$	$3\epsilon^2/8$
02	exact	exact
03	$3\epsilon^2/8$	$3\epsilon^2/8$
12	exact	exact
13	exact	exact
23	exact	exact

B. Reciprocity-closed four-photon event

$$\mathbf{r} = \mathbf{s} = (0, 1, 2, 1). \quad (23)$$

pair	X_{pq}	Y_{pq}
01	$\epsilon^2/64$	$\epsilon^2/64$
02	$\epsilon^2/4$	$\epsilon^2/4$
03	$\epsilon^2/64$	$\epsilon^2/64$
12	$\epsilon^2/64$	$25\epsilon^2/64$
13	$\epsilon^2/4$	exact
23	$\epsilon^2/64$	$25\epsilon^2/64$

For pair 13, direct evaluation at arbitrary angle gives the unnormalized coefficient $4i \sin(2\epsilon)$. Fock normalization therefore yields

$$\mathcal{A}_{X_{13}}(\epsilon) = \frac{i \sin(2\epsilon)}{4}, \quad \mathcal{A}_{Y_{13}}(\epsilon) = 0. \quad (24)$$

C. Eleven-photon affine-family root

$$\mathbf{r} = (0, 1, 3, 7), \quad \mathbf{s} = (1, 3, 3, 4). \quad (25)$$

pair	X_{pq}	Y_{pq}
01	$595\epsilon^2/8192$	$455\epsilon^2/16384$
02	$315\epsilon^2/16384$	$315\epsilon^2/8192$
03	$875\epsilon^2/16384$	$315\epsilon^2/16384$
12	$315\epsilon^2/16384$	$315\epsilon^4/8192$
13	$315\epsilon^2/16384$	$315\epsilon^2/16384$
23	$35\epsilon^2/4096$	$35\epsilon^2/4096$

For Y_{12} , the two first-neighbor permanents are equal:

$$Z_{\mathbf{r},(1,2,4,4)} = Z_{\mathbf{r},(1,4,2,4)} = 241920. \quad (26)$$

Since $s_1 = s_2 = 3$, Eq. (21) vanishes. The exact second-order common-denominator moment is

$$2903040(1 - i). \quad (27)$$

Including the Taylor factor $1/2!$ and Fock normalization gives $315\epsilon^4/8192$.

All three traceless diagonal $SU(4)$ directions preserve every tabled zero: an appended diagonal unitary multiplies the target amplitude by $\exp(i \sum_j s_j \theta_j)$.

V. PROOF OF THE ALL-ODD EXACT Y_{13} AXIS

Let $\mathbf{r} = \mathbf{s} = (0, a, 2a, a)$, with a odd. Put $x = x_1$, $y = x_2$, $z = x_3$, $u = x + z$, and $v = x - z$. For $c = \cos \epsilon$, $d = \sin \epsilon$, define $\alpha = c + d$ and $\beta = c - d$. The two mixed Fourier row forms obey

$$L'_1 L'_3 = \alpha \beta (y^2 + v^2) + i(\alpha^2 - \beta^2) y v, \quad L_2 = y - u. \quad (28)$$

The desired coefficient is proportional to

$$[x^a y^{2a} z^a] (A(y^2 + v^2) + B y v)^a (y - u)^{2a}, \quad (29)$$

where $A = \alpha\beta$ and $B = i(\alpha^2 - \beta^2)$.

The functional $[x^a z^a]$ is invariant under $x \leftrightarrow z$, which sends $v \mapsto -v$. It kills every term with an odd number of Byv factors. Choose $2h$ such factors. If $k - h$ of the remaining factors contribute v^2 , the central coefficient is

$$K_{a,k} = [x^a z^a](x - z)^{2k}(x + z)^{2a-2k}, \quad (30)$$

with weight proportional to

$$\binom{a-2h}{k-h} \binom{2a}{2k}. \quad (31)$$

The weight is invariant under $k \leftrightarrow a - k$. Substituting $z \mapsto -z$ gives

$$K_{a,a-k} = (-1)^a K_{a,k}. \quad (32)$$

For odd a , each k contribution cancels its $a - k$ partner. Hence Eq. (29) vanishes for every ϵ .

VI. NUISANCE-FLOOR AND FINITE-SHOT CALCULATION

The illustrative model in the main text is

$$P_{\text{obs}} = \mathcal{V}P_{\text{ind}} + (1 - \mathcal{V})P_{\text{dist}} + B. \quad (33)$$

For mutually distinguishable labelled photons, P_{dist} is the coefficient of $\mathbf{y}^s$ in

$$\prod_{k=0}^3 \left(\sum_{j=0}^3 |U_{jk}|^2 y_j \right)^{r_k}. \quad (34)$$

At $\epsilon = 0.1$ for the four-photon 13 comparison, the exact program gives

$$\begin{aligned} P_{\text{ind}}^X &= 0.00246684393741, & P_{\text{ind}}^Y &= 0, \\ P_{\text{dist}}^X &= 0.046875, & P_{\text{dist}}^Y &= 0.0474917109844. \end{aligned} \quad (35)$$

Substitution of $\mathcal{V} = 0.99$ and $B = 0$ gives the values in the main text. For M independent accepted trials at each setting, the difference of empirical proportions has approximate variance

$$\frac{P_X(1 - P_X) + P_Y(1 - P_Y)}{M}. \quad (36)$$

Equating the mean contrast to Z standard deviations gives the reported normal-approximation estimate. Exact binomial likelihoods should replace this estimate when counts are very small.

Uniform or mode-dependent attenuation placed strictly after the complete interferometer sends a target amplitude to

$$\mathcal{A}_{\mathbf{r},s} \prod_j \eta_j^{s_j/2} \quad (37)$$

and cannot lift a zero when the full photon number is postselected. Loss interleaved with the interferometer is different: it can change the relative weights of interfering paths. The present model does not claim to identify that loss, general partial distinguishability, source impurity, or detector backgrounds from a single observed floor.

VII. REPRODUCIBILITY

The repository uses only the Python standard library. The relevant commands are

```
python3 -m unittest discover -s tests -v
python3 scripts/analyze_reciprocity_census.py
python3 scripts/analyze_unitary_leakage.py
python3 scripts/analyze_finite_shot_protocol.py
```

The phase histograms and leakage moments are exact integer or rational calculations. Floating-point arithmetic is used only for the illustrative finite-angle nuisance model; the script separately certifies the exact $\sin^2(2\epsilon)/16$ law to numerical tolerance at multiple angles.